



Ray Tracing Program

Level 1: The Core Loop

```
For each pixel (x, y) on screen:  
    Create a ray from camera through this pixel  
    Find the closest intersection with any object  
    Calculate color at intersection point  
    Write color to pixel
```

Level 2: Essential Sub-Concepts

Each step above breaks down into smaller pieces:

1. Camera and Rays

- How do we define a camera in 3D space?
- How do we generate a ray for a given pixel?
- What's a ray mathematically? (origin + direction $\times$ t)

2. Ray-Object Intersection

- How do we know if a ray hits a sphere?
- How do we know if it hits a plane? And a cylinder?
- How do we find the closest intersection?

3. Lighting and Color

- Once we hit a point, what color should it be?
- How do lights affect the color?
- What's ambient lighting? And diffuse lighting?